

TASK SPEC AND EVALUATION RUBRIC — Delivered- Network Footprint and Connectivity (PI copy)

PI-facing companion to `2026-10-01-delivered-network-footprint-and-connectivity.md`, which is the version wavenet_junior receives. This copy carries the analytical specification in full, the reference answers, and a rubric for grading the returned work. Not for distribution to the team.

1. WHY THIS TASK EXISTS

Every coverage artifact in `metadata3/` describes the PLANNED network — the station manifest joined against a data-centre availability inventory. None of it describes the DELIVERED network. The v1 defect that cost 954 stations came from exactly that conflation, so the gap is not academic.

The task is also pitched to move the junior from monitoring to analysis: there is a defined result to produce and defend, not a status report. The script is written and verified, so failure modes are interpretive rather than mechanical — which is what makes the work gradeable.

Secondary purpose: `kappa` (resolution-ellipse aspect ratio) and the Fresnel width give a quantitative, checkable claim about the resolution this dataset supports. That is the form an external reviewer or funding agency can verify, unlike a station count.

2. ANALYTICAL SPECIFICATION

2.1 Measurement chain

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Stage 0  reconcile manifest (1,999) / result JSONs (1,933) / `.h5` on disk (1,080),
         classifying every disagreement. Catches real data loss (`ok_json_without_h5`)
         and distinguishes it from the benign in-flight case.
Stage 1  per station, read group attrs + `_coverage` bitmaps only. BH3 sbatch, ~2 min,
         emits a 240 kB cache. Never slices a channel dataset (~1.2e9 float32, 4.9 GB).
Stage 2  footprint, packaged volume, channel-band mix.
Stage 3  all-pairs shared days, ray-path rasterization, per-cell metrics, figures.
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2.2 Pair construction

Shared days for all pairs at once via boolean matrix product: with M the station-by-day coverage matrix, $M^M T$ gives concurrent recording days for every pair. Self-checked against explicit logical-AND on 20 random pairs each run (prints PASS/FAIL; hard-fails on FAIL).

Gate retained from `build\_pairs.py`: ≥ 90 shared days under 1,500 km, ≥ 365 over. Applied with NO distance band — the headline analysis reports all separations. Restriction is a sub-analysis, and every restriction level is reported in absolute terms and differenced against the unrestricted baseline, never chained against the previous row.

2.3 Grid

Equal-area Fibonacci (golden-angle) sphere, nearest-cell assignment by haversine.
Equal-area and compact; healpy is unavailable on every machine in this project.

Physical motivation: Rayleigh waves at 10–40 s with group velocity 3–4 km/s give $\lambda \sim 30\text{--}160$ km. Resolution is bounded not by λ but by the first Fresnel zone, $w = \sqrt{\lambda L / 2} \sim 224$ km for $\lambda=100$ km over $L=1,000$ km. That justifies $\sim 12,000$ cells (206 km).

BUT the delivered network currently yields only 7,575 usable paths, and at 12,000 cells the per-cell azimuth statistics are dominated by nearest-neighbour binning aliasing on the cell lattice – it renders as spiral banding that is an artifact, not structure. Default is therefore 3,000 cells (412 km): the finest grid the present path count supports. This is itself a result, and should be revisited once the long-duration in-flight stations land.

2.4 Quality, per ray path

S_{ij} concurrent recording days. NCF SNR grows as $\sqrt{\text{stacked days}}$, so S is the primary per-path quality measure, not merely a gate.

Deliberately NOT collapsed to a single normalized score. The $\sqrt{}$ scaling is sound, but the prefactor – how many days suffice – varies by close to an order of magnitude with noise source distribution and seasonality, station noise floor, distance and period band. The script reports the raw distribution and counts at a ladder of standards (90/180/365/730 d). A weighted form $q = (S/S_{\text{ref}})^p$ is available but opt-in, with S_{ref} and p printed in the caption. Settling the constant properly requires the NCF convergence experiment in §6.

2.5 Quality, per cell

n_c	path count. More is better – better averaging, room to reject data. High redundancy is desirable; nothing here recommends pruning.
ladder	n_c restricted to paths meeting each overlap standard. Divergence from raw n_c localises coverage that looks adequate but is temporally thin.
A_c	azimuthal spread. With traverse length L_k and LOCAL along-path azimuth θ_k (azimuth rotates >10 deg over 1,000 km at mid-latitude): $R_c = \sum_k L_k \exp(i 2 \theta_k) / \sum_k L_k$ $A_c = 1 - R_c$ Angle doubling makes the statistic respect that ray paths are undirected ($\theta = \theta + 180$). Without it, two identical parallel paths cancel and score as perfectly spread – the classic error.
κ_c	$(1+R)/(1-R) = 2/A - 1$. Aspect ratio of the cell's resolution ellipse.
gap	max azimuth gap mod 180. Catches A_c 's blind spot: two orthogonal families score $A=1$ but leave a 90 deg gap, and 2-psi anisotropy needs ≥ 3 orientations.
w_c	median Fresnel width over the cell's crossings, in km, directly comparable to the 412 km cell side. Converts "shorter paths are better" into a resolution length rather than an arbitrary score.
ρ_c	azimuthal redundancy, descriptive only.

Resolved := $n_c \geq 10$ AND $A_c \geq 0.6$ ($\kappa_c \leq 2.33$) AND gap ≤ 60 deg.

Thresholds are conventions, parameterized, and swept – the script reports sensitivity.

2.6 Delivered vs planned

Station recovery $f_{sta} = 1023/1999 = 0.512$ is the wrong headline: pairs scale roughly as the square of local station density. Report instead

$$\psi = f_{path} / f_{sta}^2$$

$\psi \sim 1$ losses random; $\psi < 1$ losses concentrated in dense clusters (cheap, since those pairs were largely redundant); $\psi > 1$ losses concentrated in sparse regions (expensive, since those pairs were the only ones crossing their cells).

3. REFERENCE RESULTS (PI run, 2026-10-01)

Reference output committed at ``metadata3/delivered_network_reference/``.

FOOTPRINT

manifest stations	1,999
delivered (packaged, with data)	1,023 (51.2%)
days of real data	510,501
packaged on disk	502.1 GB
median days per station	229
band mix	BH 767, LH 155, HH 80, EH 15, SH 5, MH 1

RAY PATHS (all separations)

usable station pairs	7,575 (of 522,753 geometrically possible)
median shared days per pair	493
>=90 / 180 / 365 / 730 d	7,575 / 6,882 / 6,233 / 1,779
median path length	7,752 km

ILLUMINATION (3,000 cells, 412 km)

cells with >=1 path	3,000 (100%)
cells resolved	1,490 (49.7%)
median A_c	0.607 (kappa 2.30)
median Fresnel width	831 km
Gini of paths per cell	0.352

DELIVERED vs PLANNED (110–1110 km band)

psi	0.48	-> losses fell in dense regions (cheap)
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RESTRICTION SWEEP

max km	pairs	% lit	% resolved	Fresnel km
1110	1,222	18.5	1.4	186
1500	1,642	24.8	2.3	238
2000	1,732	27.4	2.5	238
3000	1,957	36.9	2.8	270
5000	2,586	56.1	5.8	393
10000	4,862	93.2	17.1	647
none	7,575	100.0	49.7	831

TEMPORAL BIAS OF THE CURRENT SNAPSHOT

delivered	n=976	median 624	advertised days	28% with 5+ yr
still in flight	n= 53	median 3,917	advertised days	96% with 5+ yr

Defensible resolution statement supported by the above: at the current 96%–complete snapshot, the delivered network illuminates 100% of the globe at ~831 km effective resolution, and meets the resolved criterion (>=10 paths, aspect ratio <=2.33, azimuth gap <=60 deg) over 49.7% of global area. Restricting to the 110–1110 km production band instead gives ~186 km resolution over 18.5% of the globe. Both are measured, not asserted.

4. WHAT THE JUNIOR WAS ASKED FOR

Mechanical: build the axon-1 venv; sbatch the BH3 scan; pull the cache; run stages 2-3; compare against the reference run.

Interpretive, the part that is actually graded — four questions:

- Q1 Which maximum path length would you choose for a global study, and why?
- Q2 Why are resolved cells over continents and poorly resolved ones over ocean basins?
- Q3 Is the 90-day minimum overlap right, and what would you measure to settle it?
- Q4 Is the delivered network good enough to do science with yet, or do we wait?

Deliverables: `stage\_5\_delivered\_network\_coverage.md` filled in, plus a Slack post following the 8-part template.

5. EVALUATION RUBRIC

Mechanical competence — necessary, not sufficient

- [] venv built; scan submitted and completed; cache transferred; stages 2–3 run clean
- [] SUMMARY.txt reproduces the reference within campaign drift
- [] notices and explains any divergence rather than reporting it silently
- [] stage doc fields all filled; Slack post follows the template's structure and voice

Q1 — path length trade-off

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| Strong | engages the trade-off as a trade-off: short paths give ~186 km resolution over 18.5% of the globe, unrestricted gives ~831 km everywhere; picks one and ties the choice to an actual scientific target (global model vs regional detail); may note the 110 km floor exists for far-field validity. |
| Adequate | reads both curves correctly, picks one, gives a reason. |
| Weak | restates the numbers; or declares one "better" with no stated objective. |

Q2 — continental vs oceanic resolution

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| Strong | identifies that stations are overwhelmingly on land, so paths crossing an ocean basin run between continental margins and sample a narrow set of orientations; connects this to LOW A_c (anisotropic class) rather than to low path count, and notices the distinction – basins are lit but badly conditioned, not unlit. |
| Adequate | "stations are on land, so ocean coverage is worse." |
| Weak | attributes it to too few paths without checking that illumination is 100%. |

Q3 — the 90-day threshold

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| Strong | recognizes it as a convention rather than a measurement; proposes the convergence experiment (stack 30/60/.../730 days on real pairs, measure SNR vs stack length, find where it plateaus); notes it varies with distance, band and site noise, so one global constant is suspect; may point at `ncf_candidate_pairs.csv` as the input. |
| Adequate | says it should be tested empirically, without specifying how. |
| Weak | accepts 90 days because it is what the code does. |

Q4 — is it usable yet

Strong	argues from the temporal-bias evidence that 7,575 is a floor, since in-flight stations are 96% long-duration and will pair broadly; distinguishes "usable for method development now" from "usable for a publishable global model"; may note $\psi=0.48$ means remaining failures are low-priority relative to finishing.
Adequate	takes a position supported by at least one real number.
Weak	yes/no with no quantitative support.

Bonus credit

- [] independently questions a methodological choice (grid resolution, the resolved thresholds, the Fresnel lambda, the gate) with reasoning
- [] spots something in the figures that looks wrong and says so
- [] identifies a result the PI did not flag

Automatic concern — regardless of the rest

- [] writes into `packaged\_h5/` or violates the hard-link rule
- [] slices a channel dataset
- [] runs anything on terravibranium
- [] reports numbers that cannot be reproduced from their own output

6. OPEN ITEMS THIS TASK FEEDS

1. Re-run at campaign completion. The usable-pair count should rise substantially; the 12,000-cell (206 km) grid may become supportable. Cheap to repeat by design.
2. NCF convergence experiment (wavenet_senior). `ncf\_candidate\_pairs.csv` is the input. Replaces the 90-day convention with a measured number and retires the arbitrary normalization constant discussed in §2.4.
3. Median delivered duration (229 d) versus the availability inventory's median for the same stations (624 d). Inventory-methodology artifact, or genuine retrieval shortfall? Not investigated. The download window is unbounded (1970 to present), so it is not a configured cap.
4. `build\_coverage\_density\_analysis.py` carries three confirmed defects – straight lat/lon interpolation instead of great circles (also wrong across the antimeridian), a non-equal-area 2-degree grid, and a chained differential mask across a moving band edge. Its published numbers need a caveat. Fix, supersede, or annotate?
5. Mirror automation, unchanged from the previous memo: the three mirror scripts remain one-time snapshots. terravibranium is now current (1,023 files / 468 GB) following the targeted 9-file copy, but will drift again.